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UNIT → 3

→ SIGNIFICANCE OF ψ

→ $\psi(r,t)$ which is a sol^n of Schrodinger equation provides information about the behaviour of particle

→ ψ maybe regarded as measure of probability of finding particle around particular position.

→ Postulate by Max Born states that $|\psi|^2$ maybe interpretted as probability density of particle associated with De-Broglie wave described by ψ which is a complex quantity expressed as

$$\psi(r,t) = \psi(x,y,z,t) = a+ib$$

Also complex conjugate of ψ

$$\psi^*(r,t) = \psi^*(x,y,z,t) = a-ib$$

$$\Rightarrow \psi\psi^* = (a+ib)(a-ib) = a^2+b^2$$

This maybe denoted by P i.e.

$$P = |\psi(x,y,z,t)|^2 = a^2+b^2$$

→ $\psi\psi^* > 0$ if $\psi \neq 0$

→ $|\psi(x,y,z,t)|$ is modulous of ψ

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→ $|\psi(x,y,z,t)|^2$ is probablity density

→ Probablity of finding a element in volume $d\tau = dx\,dy\,dz$ at (x,y,z) is

$$P\,d\tau = \psi(x,y,z,t)\,\psi^*(x,y,z,t)\,d\tau$$

$$= |\psi(x,y,z,t)|^2\,d\tau$$

For motion of particle in 1-D

$$P\,dx = |\psi(x,t)|^2\,dx$$

It's the probablity that the particle would be found over small distance dx at posn x & time t & $|\psi(x,t)|^2$ is called probablity per unit distance i.e. position probablity density

since $\psi(x,t) = \psi(x)\,e^{-\frac{iEt}{\hbar}} = \psi(x)\,e^{-i\omega t}$

then $\psi^*(x,t) = \psi^*(x)\,e^{i\omega t}$

$$P\,dx = \psi(x,t)\,\psi^*(x,t)\,dx = |\psi(x,t)|^2\,dx$$

Illy in 3-D

$$P\,d\tau = |\psi(x,y,z)|^2\,d\tau$$

→ TIME DEPENDENT SCHRODINGER EQUATION

Let's us assume that trajectory of particle should be described by a wave function $\psi(r,t)$ whose magnitude is large in regions where probability of occurrence of particle is large.

$$\psi(r,t) = A e^{-i(\omega t - Kr)}$$

which represents the disturbance of amplitude A & of wavelength $\lambda = \frac{2\pi}{K}$ travelling in direction of wave vector K with phase velocity $\frac{\omega}{K}$

If we assume propagation along x - axis

$$\psi(x,t) = A e^{-i(\omega t - Kx)}$$

Also, since $P = \hbar K$ and $E = \hbar\omega$

$$\psi(x,t) = A e^{\frac{-i}{\hbar}[Et - px]} \quad - (1)$$

Differentiate (1) twice wrt x

$$\frac{d\psi(x,t)}{dx} = \frac{ip}{\hbar} A e^{\frac{-i}{\hbar}(Et - px)}$$

$$\Rightarrow \frac{d^2 \psi}{dx^2} = \frac{i^2 p^2}{\hbar^2} A e^{-\frac{i}{\hbar}(Et - px)}$$

Also from ①

$$\Rightarrow \frac{d^2 \psi}{dx^2} = -\frac{p^2}{\hbar^2} \psi$$

$$\Rightarrow \hbar^2 \frac{d^2 \psi}{dx^2} = -p^2 \psi \quad - ②$$

Now differentiate ① wrt t once

$$\Rightarrow \frac{d\psi(x,t)}{dt} = -\frac{iE}{\hbar} A e^{-i/\hbar (Et - px)}$$

$$\Rightarrow \frac{d\psi}{dt} = -\frac{iE}{\hbar} \psi$$

X B.S by i

$$\Rightarrow i\hbar \frac{d\psi}{dt} = E\psi \quad - ③$$

Also, for non relativistic free particle

$P.E = 0$

$\Rightarrow E = KE = \frac{p^2}{2m}$

X b.s. by ψ

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$\Rightarrow E\psi = \frac{p^2}{2m}\psi$

Put values of $p^2\psi$ & $E\psi$ from ②, ③

$$\Rightarrow i\hbar \frac{d\psi(x,t)}{dt} = \frac{-\hbar^2}{2m} \frac{d^2\psi(x,t)}{dx^2}$$

1-D time dependent schrodinger equation for non relativistic particle

$$i\hbar \frac{d\psi}{dt} = E\psi \quad \& \quad -\hbar^2 \frac{d^2\psi}{dx^2} = p^2\psi$$

→ IF IT HAS POTENTIAL ENERGY

$$\Rightarrow E = \frac{p^2}{2m} + V(x,t)$$

$$E\psi = \frac{p^2}{2m}\psi + V(x,t)\psi$$

$$\Rightarrow i\hbar \frac{d\psi(x,t)}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi(x,t)}{dx^2} + V(x,t)\,\psi(x,t)$$

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→ FOR 3-DIMENSION

$$i\hbar \frac{d\psi}{dt} = \frac{-\hbar^2}{2m}\left(\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2}\right) + V\psi$$

where ψ is wave func of (x, y, z, t)

$V = V(x, y, z, t)$

Laplacian ↑ operator

also,

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} = \nabla^2\psi$$

$$\Rightarrow i\hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m}\nabla^2\psi + V(r,t)\psi$$

It's generally expressed as

$$i\hbar \frac{d\psi_{(x,t)}}{dt} = H\psi_{(x,t)}$$

H → Hamiltonian

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→ Particle in a box / ∞ square well

It describes a particle that's confined between 2 rigid & impenetrable walls. Outside box, the potential energy is ∞ (meaning particle can't exist there) & inside it's zero so the particle is free to move but only inside box.

For 1D box of length L, potential energy $V(x)$ is defined as:

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & x \le 0 \text{ or } x \ge L \end{cases}$$

By schrodinger Eq

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Inside box $V(x) = 0$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

$$\Rightarrow \frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

Let $K^2 = \frac{2mE}{\hbar^2}$

$$\Rightarrow \frac{d^2\psi(x)}{dx^2} + K^2\psi(x) = 0$$

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The general sol^n of this diff eq is :-
$\psi(x) = A \sin Kx + B \cos Kx$

A & B are constants determined from the boundary conditions i.e. $\psi(0) = 0$, $\psi(L) = 0$

1) At $x = 0$
$\psi(0) = A \sin 0 + B \cos 0$
$0 = B$
$\Rightarrow B = 0$

11) At $x = L$
$0 = A \sin KL$
$A \neq 0 \Rightarrow \sin KL = 0$
$\Rightarrow KL = n\pi$ for $n = 1, 2, 3$
$\Rightarrow K = \frac{n\pi}{L}$

$\Rightarrow$ $\psi_n(x) = A \sin\left(\frac{n\pi x}{L}\right)$

Also
$$\int_0^L |\psi_n(x)|^2 \, dx = 1$$

$$\Rightarrow A^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1$$

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$$\Rightarrow \frac{A^2}{2}\int_0^L 1 - \cos\left(\frac{2n\pi x}{L}\right) dx = 1$$

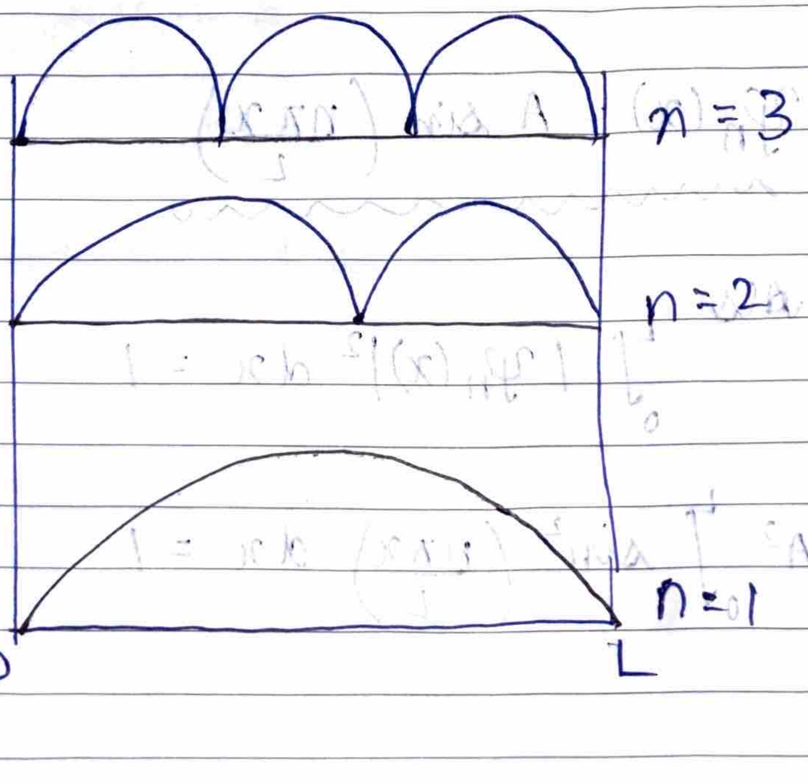
On solving, $A = \sqrt{\frac{2}{L}}$

$$\Rightarrow \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

$\psi_n(x)$ are sine waves fitting inside box

Nodes → zero pts.

n	No of nodes	Shape
1	0	1 hump
2	1	2 hump
3	2	3 hump.



→ Time independent schrodinger equation
From time dependent schrodinger equation, we have

$$① - i\hbar \frac{d\psi(x,t)}{dt} = H\psi_{(x,t)} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(x,t)\right]\psi_{(x,t)}$$

Assuming a separable solution

$$\psi_{(x,t)} = \psi(x)\,T(t)$$

Substitute in ①

$$\Rightarrow i\hbar\,\psi(x)\frac{dT(t)}{dt} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(x)\right]\psi(x)\,T(t)$$

Divide both sides by $\psi(x)\,T(t)$

$$\Rightarrow i\hbar\frac{1}{T(t)}\frac{dT(t)}{dt} = \frac{1}{\psi(x)}\left[-\frac{\hbar^2}{2m}\nabla^2 + V(x)\right]\psi(x)$$

Left side depends completely on t and right side on x therefore both must be equal to same constants. denote that value by E (energy eigen value)

$$\Leftrightarrow \frac{i\hbar}{T}\frac{dT}{dt} = E \quad \& \quad \left[-\frac{\hbar^2}{2m}\nabla^2 + V(x)\right]\psi(x) = E\psi(x)$$

$$\Rightarrow i\hbar\frac{dT}{dt} = ET$$

$$\Rightarrow T(t) = e^{-iEt/\hbar} \quad \& \quad E\psi(x) = \left[\frac{-\hbar^2}{2m}\nabla^2 + V(x)\right]\psi(x)$$

→ Normalisation of wave functions
$\psi(x,t)$ must give a meaningful probablity interpretation hence probablity of finding the particle in complete space must be equal to 1

$$\Rightarrow \int_{-\infty}^{\infty} \underbrace{|\psi(x,t)|^2}_{\text{Represents Probablity Density}} dx = 1$$

Normalisation conditions

→ Properties of a well behaved wave function
1. Must be single valued
At any pt. x, ψ must have only 1 value
2. Must be continous
No sudden jumps are allowed in ψ because infinite jumps give ∞ kinetic energy
3. First derivative must be continous
$\frac{d\psi}{dx}$ → continous except at a point of ∞ potential
4. Must be finite
ψ must not diverge to ∞
5. Must be square integrable
$\int |\psi^2| dx < \infty$
6. Must be normalisable
$\int |\psi|^2 dx = 1$ can be achieved

7. Must vanish at ∞
at $x = \infty$, $\psi(x) = 0$

→ Probablity Density

$$P(x,t) = |\psi(x,t)|^2 = \psi(x,t)\,\psi^*(x,t)$$

$P(x,t)\,dx$ is probablity of finding particle between x & $x + dx$

$$\int_{-\infty}^{\infty} |\psi(x,t)|^2\,dx = 1$$

→ Probablity current density $J(x,t)$

It describes how probablity flows in space

$$J(x,t) = \frac{\hbar}{2mi}\left[\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx}\right]$$

Also,

$$\left.\frac{dP}{dt} + \frac{dJ}{dx} = 0\ \right\}$$ Total probablity is conserved

current density is non zero for travelling waves & zero for bound states.

→ expectation values

For an operator $\hat{A}$, the expectation value is

$$\langle A \rangle = \int_{-\infty}^{\infty} \psi^*(x,t)\,\hat{A}\,\psi(x,t)\,dx$$

→ $\langle x \rangle = \int x\,|\psi|^2 dx$ $\quad [\hat{x} = x]$

→ $\langle p \rangle = \int \psi^* \left(-i\hbar \frac{d}{dx}\psi\right) dx$ $\quad \left[\hat{p} = -i\hbar \frac{d}{dx}\right]$

→ energy

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$$

$$\langle H \rangle = \int \psi^* \hat{H} \psi \, dx$$